Seminar5

luni, 8 noiembrie 2021 10:03

Lema de pompare

Dacă $L \in LR$ atunci $\exists\, n_0 \in \mathbb{N}^*$, $\forall\, w \in L$, $|w| > n_0$, $\exists$ un mod de a împărți cuvântul w în $x, y, t \in \Sigma^*$ cu proprietățile că $w = xyt$, $|y| > 0$ și $|xy| < n_0$, atunci $\forall\, k \in \mathbb{N}$ $xy^k t \in L$.

$a^* bc$

$x = a$, $y = a$, $t = bc$

$xy^k t = a a^k bc = a^{k+1} bc \in L$, $\forall\, k \in \mathbb{N}$

$x = e$, $y = aa$, $t = bc$

$xy^k t = e \cdot (aa)^k bc = (aa)^k bc \in L$, $\forall k$

Dacă L nu respectă lema de pompare atunci $L \notin LR$.

$a^n b^n$

I. $x = a^p$, $y = a^2$, $t = a^{n-p-2} b^n$

II. $x = a^p$, $y = a^{n-p}$, $t = b^n$

III. $x = a^p$, $y = a^{n-p} b^2$, $t = b^{n-2}$

IV. $x = a^m$

$y = b^p$

$z = b^{m-p}$

V. $x = a^m b^p$

$y = b^2$

$z = b^{m-p-2}$

I. $x = a^p$

$y = a^2$

$z = a^{m-p-2} b^n$

$(a^2)^3 = a^6$

$(aa)(aa)(aa) = aaaaaa$

$xy^k z = a^p (a^2)^k a^{m-p-2} b^n = a^{p+2k+m-p-2} b^m$

$\Rightarrow \cancel{p} + 2k + m \cancel{-p} - 2 = m \quad , \forall \; k \in \mathbb{N}$

$\cancel{m} + 2(k-1) = \cancel{m} \quad , \forall \; k \in \mathbb{N}$

$2\underbrace{(k-1)}_{\neq 0} = \underbrace{0}_{=0} \quad , \forall \; k \in \mathbb{N}$

$\Rightarrow xy^k z \notin L.$

II. $x = a^p$

$y = a^{m-p}$

$z = b^m$

$xy^k z = a^p (a^{m-p})^k b^n = a^p a^{(m-p)k} b^m$

$= a^{p + mk - pk} b^n \in L. \Rightarrow$

$p + mk - pk = m \quad , \forall \; k \in \mathbb{N}$

$\Rightarrow xy^k z \notin L.$

III. $x = a^p$

III. $x = a^p$

$y = a^{n-p} b^2$

$z = b^{n-2}$

$xy^k z = a^p \left(a^{n-p} b^2\right)^k b^{n-2} \notin L.$

$\downarrow$

$\underbrace{a^{n-p} b^2 \, a^{n-p} b^2 \cdots a^{n-p} b^2}_{\text{de } k \text{ ori}}$

IV. $x = a^m$

$y = b^p$

$z = b^{m-p}$

$xy^k z = a^m b^{p \cdot k + m - p} \Rightarrow m = m + p(k-1), \forall k \in \mathbb{N}$

$0 = p(k-1), \forall k \in \mathbb{N}$

fals $\Rightarrow xy^k z \notin L.$

V. $x = a^m b^p$

$y = b^2$

$z = b^{n-p-2}$

$xy^k z = a^m b^{p + k2 + n - p - 2}$

$\Rightarrow m = m + 2(k-1), \forall k \in \mathbb{N}$

$0 = 2(k-1)$

fals $\Rightarrow xy^k z \notin L.$

Pt. cō ~~$\exists$~~ nicio împărțire a lui w în x, y, z

Pt. că ~~$\exists$~~ nicio împărțire a lui x^k în x, y, z

pt. care $xy^kz \in L \Rightarrow L$ nu respectă L.P.

$\Rightarrow L \notin LR$

Dc. $L_1, L_2 \in LR$ at. $L_1 \cup L_2$, $L_1 \cap L_2$, $L_1 \setminus L_2$, $L_1 \circ L_2$, CL_1, L_1^* $\in LR$.

2. Fie $L_1, L_2 \in LR$, $L_3 \notin LR$, L_4 finit.

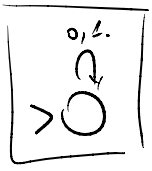
$L_5 = \left(\left(L_1 \cap L_2 \right) \cup \left(L_4 \setminus L_3 \right) \right)^* \circ \left(L_3 \cap \emptyset \right)^*$

$L_1 \cap L_2 \in LR$; $L_4 \setminus L_3 \in LR$ (finit); $\emptyset \in LR$; $\emptyset^* = e$, $e \in LR$; $\left(L_1 \cap L_2 \right) \cup \left(L_4 \setminus L_3 \right) \in LR$; $\left(\left(L_1 \cap L_2 \right) \cup \left(L_4 \setminus L_3 \right) \right)^* \in LR$; $\left(L_3 \cap \emptyset \right)^* \in LR$; $L_5 \in LR$.

$L_1, L_2 \in LR \Rightarrow L_1 \cap L_2 \in LR$ cf. prop. de închidere

$L_4 \setminus L_3$ e finit. pt. că L_4 finit.

2. Explicați cum puteți demonstra că un limbaj regulat L primit ca parametru într-o funcție este $= \emptyset$.



$\Sigma = \{0, 1\}$

$\Rightarrow \emptyset$

$E_1 = \emptyset$

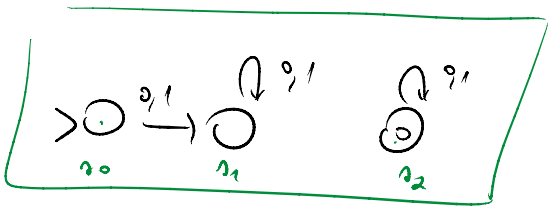
$E_2 = a$

$0 \cup 1$

$(a \cup b)^*$

$E = a^* \emptyset$

Dc. $L \in LR$. at. $\exists$ ER / AFD / AFN p. L.



Fac o parcurgere BFS pornind din q_0.

Dc. $\exists$ o cale de la q_0 la q_f, $q_f \in F$ at.

$L = \emptyset$.

2. Primim un $L \in LR$. Dem. că $L \neq \Sigma^*$.

$$C\Sigma^* = \emptyset$$

$L \neq \Sigma^* \Leftarrow | \; CL \neq \emptyset$

$\exists w \in \Sigma^*, w \notin L \Rightarrow CL = \Sigma^* \setminus L = \{w\}$

$$CL = \Sigma^* \setminus L$$

$$C\Sigma^* = \emptyset$$

$$C\emptyset = \Sigma^*$$

$$\Sigma^* \cup L = \Sigma^*$$

$$\Sigma^* \cap L = L$$

$$\emptyset \cup L = L$$

$\emptyset \cup L = L$

$\emptyset \cap L = \emptyset$

$L \cdot \emptyset = \emptyset \cdot L = \emptyset$

$L_1 \circ L_2 = \{ w \in \Sigma^* / \; w = w_1 w_2, \; w_1 \in L_1 \text{ și } w_2 \in L_2 \}$

$\emptyset^* = e$

$L \cdot e = L = e \cdot L = L$

Avem 3 AFN M_1, M_2, M_3.

m. stări	s_1	s_2	s_3
m. de tranziții	t_1	t_2	t_3

$M_1 = M_2 = M_3$

$\mathcal{L}(M_3) = \mathcal{L}(M_1) \cup \mathcal{L}(M_2)$

a) ~~$s_3 > s_1$ și $s_3 > s_2$~~

b) ~~$t_3 \geq (t_1 + t_2)$~~

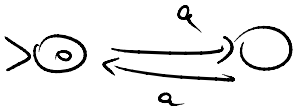
c) ~~$s_3 \geq s_1 + s_2$~~ $\vee$ $(t_3 > t_1 \wedge t_3 > t_2)$

d) mărimea unei variabile de mai mult.

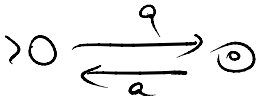
L

$\mathcal{L}(M)$

$\mathcal{L}(E)$

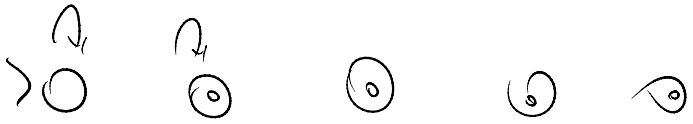
M_1 

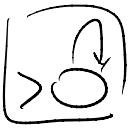
$\Sigma = \{ a \}$

M_2 

a^*

M_3 





Care din următoarele expresii sunt echivalente cu a^*b^* ?

a) $(a^*b^*)^*$ $\quad \emptyset$ $\quad$ a b c b ...

b) ✓ $a^*b^* \cup b^*a^*\emptyset = a^*b^* \cup \emptyset = a^*b^*$

c) $a^*b^* \cap \Sigma^* = a^*b^*$ * nu e exp. regulată

d) $(a^* \cup b^*)^*$ $\quad a^*b^* \cap (a \cup b)^*$

=

e

$b^*a^*\emptyset^* = b^*a^*$

$E = a^*b^*$

$E = a^*$

$\mathcal{L}(E) = \{ \varepsilon, a, aa, \ldots \}$

$E = \emptyset \qquad \Sigma^*$

$L = \Sigma^* = \{ w \in \Sigma^* \}$

$(a^* \cup b^*)(a^* \cup b^*) = b^* a^*$

$(a^* \cup b^*)^* \leq$

Fie $L_1 = \{ a^p / p \text{ este un număr prim} \} = \{aa, aaa, \ldots\}$

$L_2 = \{ a^{2k} / k \in \mathbb{N} \} \qquad (aa)^*$

$L_3 = (L_1 \cap L_2)^* \setminus L_2$ este:

a) L_1^*

b) CL_2

c) $\emptyset$

d) $L_1^* \cap L_2^*$

$L_1 \cap L_2 = \{aa\}.$

$(L_1 \cap L_2)^* = (aa)^*$

$L_2 = (aa)^* = \emptyset$

$(L_1^*)^* = L_1^*$

$$(L^*)^* = L^*$$